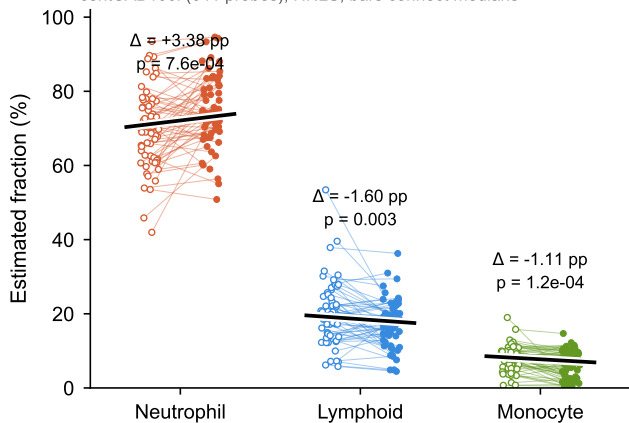


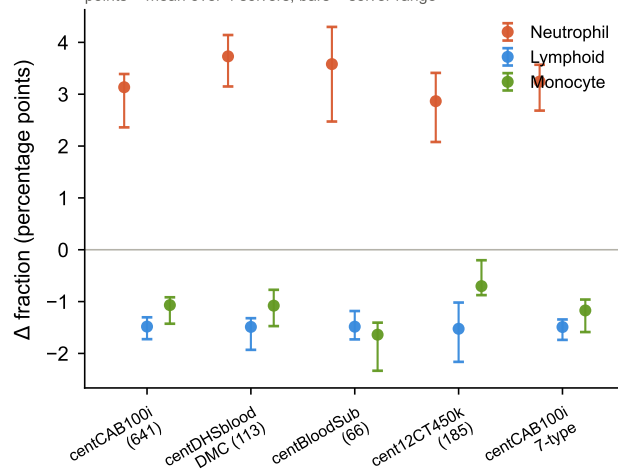
### A Paired pre → post lineage shift (n = 65)

centCAB100i (641 probes), NNLS; bars connect medians



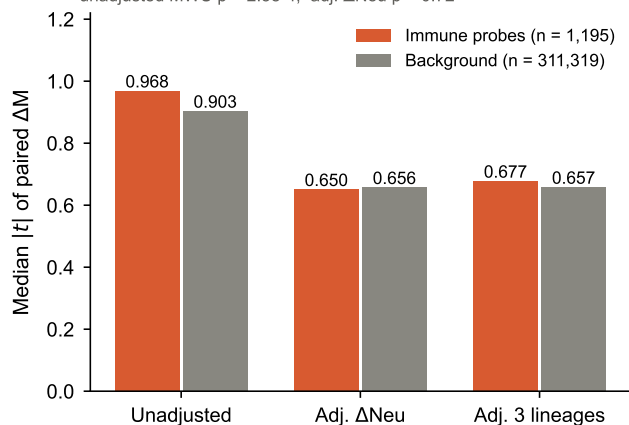
### B Shift is consistent across five references

points = mean over 4 solvers; bars = solver range



### C Probe level: immune enrichment is abolished

unadjusted MWU  $p = 2.5e-4$ ; adj.  $\Delta$ Neu  $p = 0.72$



### D DMR level: attenuated but not abolished

